



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Riverbrook 9; Beltan Properties
Location: Gainesville, GA
Description: Tranquility - UBSmt, SW-Facing, ACH50

Date: March 17, 2024
By: JTH / TJH
House Faces: SouthWest

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,741 ft ²
Volume	25,619 ft ³
Percentage Glass	14 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	12124
SEER	14
ACH50	3.0
Mechanical Ventilation	YES
House Outside Air CFM	75

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.048	Basement	Concrete + 2x6	R-19 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 sprayed	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.077	Framed		R-13 sprayed	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.30	Blinds/Draperies	SHGC 0.18
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			30473 Btu/hr	43310 Btu/hr	
Minimum Nominal Capacity			3.39 Tons	12.69 kW	
Load Density			810 SF/Ton	4.6 W/SF	
Floor Area Served			2741.0 SF	2741.0 SF	
Mechanical Ventilation			75 CFM	75 CFM	
System Size			3.5 Tons	15.0 kW	
Leaving Air Temperature			57.2 deg F	103.9 deg F	
Installed Capacity			783 SF/Ton	5.5 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1				
Size	3.50	Tons		
Airflow	1400	CFM		
Heat kW	15	kW		
First Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer / Hall	53	73	63	
Master WIC	82	95	89	
Master Bath	82	92	87	
Master Suite	121	146	133	
Family	236	211	224	
Kitchen	152	44	98	
Dining	110	167	138	
Laundry	57	32	45	
Garage Entry Hall	44	60	52	
Bath 2	26	25	26	
Bed 2 / Study	91	100	95	
-	-	-	-	
-	-	-	-	
-	-	-	-	

[illegible]

Cooling Size	2.5	DX Cooling kWh	3059
Heating Size	3	RC Heating kWh	6026
Best Case	2.5	Total kWh	9086

[illegible]

Cooling Size	1	DX Cooling kWh	1004
Heating Size	1	RC Heating kWh	2035
Best Case	1	Total kWh	3039